Business Management Study Guide

s 7 to 12 | Concepts, MCQs, and Online Assessment Questions | Written in simple language with examples

Weeks 1 to 6

Concepts explained in simple language

This PDF is arranged week by week. Each week contains a clear Concepts section, a full MCQs section, and a separate Online Assessment Questions section for extra practice.

Week 7

Concepts

1. Decision-Making Levels in Organizations

Operational management deals with daily routine decisions such as payroll processing, transaction handling, and attendance recording. These decisions are repetitive and usually follow fixed rules.

Tactical management deals with medium-term planning such as campaign design, budgeting, and departmental strategy. These decisions use both rules and human judgment.

Strategic management focuses on long-term issues such as expansion, investment, and major policy decisions. These decisions are highly unstructured and require experience, judgment, and analysis.

2. Types of Decisions

Structured decisions have clear rules and can be repeated easily. For example, approving a standard reorder point in inventory can be structured.

Semi-structured decisions use both data and human interpretation. A marketing strategy often falls into this category.

Unstructured decisions do not have a clear formula. They require reasoning, judgment, and contextual understanding, such as approving a long-term capital investment.

3. BI Infrastructure and AI-Based Decision Support

A data warehouse is the core BI infrastructure component that stores organized data for reporting, dashboards, and analytics.

AI-based decision systems work best when the objective is clear, the constraints are known, and the possible alternatives are defined. When these conditions exist, AI can support or automate decisions more reliably.

An AI assistant usually responds to a user, explains information, or generates content on request. An AI agent goes further by autonomously executing tasks, selecting tools, and completing parts of a workflow.

4. Explainable AI and Reasoning Quality

Explainable AI (XAI) helps people understand how a model reached a decision.

SHAP and LIME are popular XAI tools. They explain a prediction by showing which features pushed the result upward or downward.

This is important because decision systems are more trustworthy when people can understand the reason behind the output.

5. Correlation, Causation, and Bias

Correlation means two things move together. Causation means one actually causes the other. Analysts must not confuse the two.

Anchoring bias happens when the first piece of information influences judgment too strongly. Confirmation bias happens when people only look for evidence that supports what they already believe. Overconfidence bias happens when people trust their own judgment too much.

6. Decision Trees

A decision tree is a supervised learning model that makes predictions through a series of if-then splits. Its goal is to create groups that are as pure or homogeneous as possible.

The root node is the starting point and represents the full dataset. A split divides the data based on a selected attribute. A leaf node gives the final prediction or class.

Induction means learning general rules from data examples.

7. Attribute Selection and Tree Growth

To choose the best attribute for a split, decision trees use measures such as Gini Index and Entropy. Lower impurity means a better split.

A greedy algorithm chooses the best local split at each step rather than searching for the globally perfect tree.

Overfitting happens when a tree becomes too detailed and memorizes the training data.

8. Pruning and Ensemble Methods

Pre-pruning stops the tree early, but it may miss useful splits. Post-pruning grows the tree fully and then cuts unnecessary branches, which is often better.

Bagging builds many trees in parallel and combines them by voting. Boosting builds trees sequentially, where each new model tries to correct the previous model?s mistakes.

Random Forest is a bagging-based method that also adds random feature selection. Bootstrapping means sampling with replacement to create many training datasets.

> *Example: In fraud detection, a decision tree may first split on transaction amount, then on country, and then on device type. Each split tries to make the fraud and non-fraud groups cleaner.*

MCQs

1. Which management level mainly handles routine daily decisions?

A) Strategic management
B) Tactical management
C) Operational management
D) Governance management

Answer: C

---

2. Which of the following is an example of an unstructured decision?

A) Payroll processing
B) Inventory reorder
C) Capital investment approval
D) Sales entry

Answer: C

---

3. Which of the following is a semi-structured decision?

A) Attendance logging
B) Marketing strategy design
C) Billing entry
D) Data entry

Answer: B

---

4. What is the core storage component in BI infrastructure?

A) Dashboard
B) Data warehouse
C) CRM
D) ERP

Answer: B

5. What is the main purpose of SHAP and LIME?

A) To train a model
B) To explain predictions
C) To clean data
D) To store data

Answer: B

6. AI-based decision systems perform best when which condition is satisfied?

A) Data is missing
B) Goals are unclear
C) Objectives are clearly defined
D) Decisions are random

Answer: C

7. What is the key feature of an AI agent?

A) Reporting only
B) Chatting only
C) Autonomous execution of tasks
D) Visualization only

Answer: C

8. What does correlation mean?

A) Cause-and-effect relationship
B) Relationship only
C) No relationship
D) Exact prediction

Answer: B

9. What does anchoring bias mean?

A) Ignoring all data
B) Letting the first information dominate judgment
C) Being too confident
D) Thinking randomly

Answer: B

10. Which type of learning do decision trees usually belong to?

A) Unsupervised learning
B) Supervised learning
C) Reinforcement learning
D) Random learning

Answer: B

11. What does the root node in a decision tree represent?

A) Final output only
B) The full dataset
C) The error rate
D) One feature value

Answer: B

12. What does a leaf node provide in a decision tree?

A) Input data
B) Final prediction
C) Data cleaning
D) Error correction

Answer: B

---

13. What is the main purpose of splitting in a decision tree?

A) To increase impurity
B) To increase purity
C) To reduce data size only
D) To add noise

Answer: B

---

14. What does homogeneity mean in a decision tree node?

A) Mixed classes
B) Mostly the same class
C) Random data
D) Missing values only

Answer: B

---

15. What is the Gini value of a completely pure node?

A) 1
B) 0
C) 0.5
D) 2

Answer: B

---

16. What is the entropy value of a completely pure node?

A) 1
B) 0
C) 0.5
D) -1

Answer: B

---

17. What does a lower Gini value indicate?

A) A worse split
B) A better split
C) No split
D) Random behavior

Answer: B

---

18. What type of search strategy do decision trees typically use?

A) Greedy strategy
B) Random strategy
C) Linear strategy
D) Static strategy

Answer: A

---

19. In machine learning, what does greedy mean?

A) Choosing the best global option
B) Choosing the best local option at each step
C) Choosing randomly
D) Choosing the worst case

Answer: B

---

20. What does overfitting mean in a decision tree?

A) Strong generalization
B) Memorizing training data
C) Simplicity
D) Perfect real-world accuracy

Answer: B

---

21. What is a major issue with pre-pruning?

A) It always causes overfitting.
B) It may stop before finding good splits.
C) It is too fast.
D) It increases complexity.

Answer: B

---

22. Which pruning method is usually considered better?

A) Pre-pruning
B) Post-pruning
C) No pruning
D) Random pruning

Answer: B

---

23. What does cost-complexity pruning mainly penalize?

A) Accuracy
B) Complexity
C) Data volume
D) Number of classes

Answer: B

---

24. What does the term ensemble mean in machine learning?

A) One model only
B) Multiple models together
C) No model
D) A linear model

Answer: B

---

25. How are models built in bagging?

A) Sequentially
B) In parallel
C) Randomly without training
D) In a linear chain only

Answer: B

---

26. How are models built in boosting?

A) In parallel
B) Sequentially
C) Without any order
D) In a static way

Answer: B

---

27. What additional idea does Random Forest introduce?

A) Noise injection only
B) Random feature selection
C) Error deletion
D) High bias only

Answer: B

---

28. What does bootstrapping mean?

A) No sampling
B) Sampling with replacement
C) Sampling without replacement
D) Data cleaning

Answer: B

---

29. What kind of voting is commonly used in bagging classification?

A) Weighted voting only
B) Majority voting
C) Random voting
D) No voting

Answer: B

---

30. What does boosting mainly improve?

A) Randomness
B) Error correction across models
C) Noise only
D) Data size

Answer: B

---

31. What shape is commonly used to represent a decision tree?

A) Matrix
B) Inverted tree structure
C) Circle
D) Histogram

Answer: B

---

32. Does a decision tree classification problem require a target variable?

A) Yes
B) No
C) Only sometimes in unsupervised learning
D) Only in clustering

Answer: A

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33. What is the main use of a data warehouse?

A) Game development
B) Analysis and reporting
C) Image generation
D) Hardware control

Answer: B

---

34. What is the main purpose of XAI?

A) Faster training
B) Transparency
C) Higher storage
D) Lower memory

Answer: B

---

35. What is the main difference between an AI agent and an AI assistant?

A) Interface color
B) Accuracy
C) Autonomy
D) Model size

Answer: C

---

36. What is the common mistake people make with correlation?

A) Assuming it implies causation
B) Ignoring variance
C) Confusing it with entropy
D) Treating it as noise only

Answer: A

---

37. Which of the following is a cognitive bias?

A) Anchoring bias
B) Gini bias
C) Entropy bias
D) Bootstrap bias

Answer: A

---

38. What defines a good split in a decision tree?

A) Higher impurity
B) Larger dataset only
C) Lower impurity
D) Longer runtime

Answer: C

---

39. Why is node size important in a decision tree?

A) It affects whether the split is practical and actionable.
B) It only changes color in diagrams.
C) It matters only for storage.
D) It has no effect.

Answer: A

---

40. What is the maximum entropy of a binary split?

A) 0
B) 0.25
C) 0.5
D) 1

Answer: D

---

41. What is the maximum Gini impurity of a binary split?

A) 0
B) 0.25
C) 0.5
D) 1

Answer: C

---

42. What does induction mean in a decision tree context?

A) Learning patterns from data
B) Deleting data
C) Sorting features alphabetically
D) Standardizing variables

Answer: A

---

43. What does a leaf node usually represent?

A) A final rule or prediction
B) The first split only
C) A feature list
D) A missing value

Answer: A

---

44. Which technique is commonly used to reduce overfitting in decision trees?

A) Scaling
B) Tokenization
C) Pruning
D) Clustering

Answer: C

---

45. What is one important benefit of ensemble methods?

A) Lower stability
B) Better stability and robustness
C) Guaranteed perfect accuracy
D) No training needed

Answer: B

---

46. What does bootstrapping create for bagging methods?

A) More labels
B) More bias
C) Less data
D) Diversity across training samples

Answer: D

---

47. Random Forest is often preferred because it improves what?

A) Accuracy and robustness
B) File storage only
C) Manual rules only
D) Data entry speed only

Answer: A

---

48. In which area are decision trees commonly used?

A) Fraud detection
B) Painting
C) Circuit wiring only
D) Typing practice

Answer: A

---

49. What must be clear for an AI-based decision system to work effectively?

A) Constraints and objectives
B) Colors of the dashboard
C) Random alternatives only
D) File extensions only

Answer: A

---

50. Which of the following is an example of anchoring bias?

A) Assuming a correlation proves causation
B) Ignoring all data
C) Using entropy to select a split
D) Letting the first quoted price strongly influence the final judgment

Answer: D

Online Assessment Questions

1. Which management level mainly handles routine decisions?

A) Strategic management
B) Tactical management
C) Operational management
D) Executive governance

Answer: C) Operational management

---

2. Which of the following is an example of an unstructured decision?

A) Payroll approval
B) Sales entry
C) Approving long-term capital investment
D) Inventory counting

Answer: C) Approving long-term capital investment

---

3. Which of the following is an example of a semi-structured decision?

A) Designing a marketing strategy
B) Entering transactions
C) Attendance logging
D) Printing bills

Answer: A) Designing a marketing strategy

---

4. Which BI component acts as the main central storage system for analysis?

A) Spreadsheet
B) Data warehouse
C) Email server
D) Browser cache

Answer: B) Data warehouse

---

5. What do SHAP and LIME mainly do?

A) Explain predictions using feature contributions
B) Build dashboards only
C) Remove missing values
D) Increase storage speed

Answer: A) Explain predictions using feature contributions

---

6. Under what condition do AI-based decision systems work best?

A) When goals are vague
B) When objectives, constraints, and alternatives are clearly specified
C) When training data is missing
D) When judgment is fully random

Answer: B) Objectives, constraints, and alternatives clearly specified

---

7. Which example best describes an AI agent?

A) A chatbot answering one question
B) A dashboard showing reports
C) A system autonomously designing workflows and executing tasks
D) A static FAQ page

Answer: C) Autonomously designing workflows and executing tasks

---

8. Which statement correctly describes correlation?

A) It always proves causation.
B) It means the relationship is correlational, not automatically causal.
C) It means there is no relationship.
D) It means one factor is always the reason for the other.

Answer: B) Relationship is correlational, not causal

---

9. Which bias is shown when the first information influences later judgment too strongly?

A) Confirmation bias
B) Overconfidence bias
C) Anchoring bias
D) Outcome bias

Answer: C) Anchoring bias

---

10. Which approach is often preferred when transparency is important in AI systems?

A) Use a simpler model
B) Increase hidden layers only
C) Ignore explanations
D) Maximize randomness

Answer: A) Use simpler model (transparency)

Week 8

Concepts

1. Decision Trees in Python Workflows

A decision tree is a flowchart-style predictive model. Each split tries to make the resulting groups more pure, meaning the observations inside each group become more similar in class.

A targeting problem means selecting the right customers, cases, or stores when resources are limited. The goal is to maximize results with a limited budget.

A pandas DataFrame is a table-like structure with rows and columns. It is commonly used to prepare and analyze data in Python.

scikit-learn is a Python library used to train models such as Decision Trees and Random Forests.

2. Training Data and Features

Train-test split divides the data into a training part and a testing part. The model learns patterns on the training data and is evaluated on the test data to see whether it generalizes.

Features (X) are the input variables, while target (Y) is the output the model tries to predict.

DecisionTree in scikit-learn generally expects numerical data, so text categories must be encoded or removed first.

Ordinal encoding is used when category order matters. For example, low, medium, and high can be converted to 0, 1, and 2.

3. Hyperparameters, Impurity, and Pruning

Hyperparameters are settings chosen before training. Examples include ccp_alpha and min_samples_leaf.

Gini impurity measures how mixed a node is. A value of 0 means the node is pure.

ccp_alpha controls pruning. A value of 0 means no pruning, while higher values create a smaller and simpler tree.

Overfitting happens when a tree memorizes the training data and performs poorly on unseen data.

.fit() is the method used to train the model.

4. Model Evaluation

A confusion matrix compares actual results with predicted results.

True Positive means a positive case was predicted correctly. True Negative means a negative case was predicted correctly. False Positive is a false alarm. False Negative is a missed positive case.

Recall or sensitivity measures how many actual positives were found. Specificity measures how well the model avoids false positives. Precision measures how many predicted positives are actually correct.

ROC curve plots True Positive Rate against False Positive Rate. AUC summarizes ROC performance, where 0.5 means random and 1.0 means perfect.

PR curve plots precision against recall and is especially useful for imbalanced data.

5. Imbalanced Data and Multi-Class Handling

Imbalanced data means one class is much larger than another. This can make the model biased toward the majority class.

SMOTE creates synthetic minority-class examples to balance the training data. It should be applied only to the training set, not to the test set.

LabelBinarizer converts multi-class targets into multiple binary forms so metrics can be computed more easily.

Micro averaging gives equal importance to all predictions. Macro averaging gives equal importance to all classes.

6. Random Forest and Cross-Validation

Random Forest combines many trees to improve stability and reduce sensitivity to one specific split.

Cross-validation uses repeated train-test splits to produce a more reliable estimate of model performance.

> *Example: If the dataset has 99 non-fraud cases and 1 fraud case, a model may look accurate by always predicting non-fraud. That is why recall, precision, PR curves, and SMOTE become important.*

MCQs

1. What is the main goal of a decision tree?

A) To store data
B) To create rules for prediction
C) To sort data only
D) To visualize data only

Answer: B

---

2. What does a Gini impurity value of 0 mean?

A) Mixed data
B) A pure node
C) A random split
D) An error state

Answer: B

---

3. Why is train-test split used?

A) To increase data size
B) To avoid overfitting and test generalization
C) To remove all noise
D) To encode data

Answer: B

---

4. Which of the following is a hyperparameter?

A) y_train
B) ccp_alpha
C) Prediction result
D) Accuracy score

Answer: B

---

5. What does ccp_alpha = 0 mean?

A) Fully pruned tree
B) No pruning
C) No training
D) No split allowed

Answer: B

---

6. What does a false positive mean?

A) A wrong negative prediction
B) A wrong positive prediction
C) A correct positive prediction
D) A correct negative prediction

Answer: B

---

7. What does recall measure?

A) Accuracy only
B) How many true positives are found
C) How many true negatives are found
D) Total error count

Answer: B

---

8. What does specificity focus on?

A) True positives
B) True negatives
C) False negatives only
D) Precision only

Answer: B

---

9. What is the formula for precision?

A) TP / (TP + FN)
B) TP / (TP + FP)
C) TN / (TN + FP)
D) FN / (TP + TN)

Answer: B

---

10. What does an AUC of 0.5 indicate?

A) Perfect performance
B) Random performance
C) Overfitting
D) A coding error

Answer: B

---

11. What is SMOTE used for?

A) Encoding categories
B) Balancing imbalanced data
C) Splitting data
D) Training the final model only

Answer: B

---

12. On which data should SMOTE be applied?

A) The test set only
B) The training set only
C) The whole dataset at once
D) The target column only

Answer: B

---

13. For what type of categories is ordinal encoding most suitable?

A) Unordered categories only
B) Ordered categories
C) Images
D) Pure numeric variables

Answer: B

---

14. What is a Random Forest?

A) A single decision tree
B) A collection of many trees
C) A linear model
D) A clustering algorithm

Answer: B

---

15. What does the ROC curve plot?

A) Precision against Recall
B) True Positive Rate against False Positive Rate
C) Accuracy against Loss
D) Error against Time

Answer: B

---

16. For which type of dataset is the PR curve especially useful?

A) Balanced data only
B) Imbalanced data
C) Small data only
D) Image data only

Answer: B

---

17. What does overfitting mean?

A) A model is too simple.
B) A model memorizes training data.
C) A model has no training.
D) A model always generalizes perfectly.

Answer: B

---

18. What does underfitting mean?

A) The model is too complex.
B) The model is too simple.
C) The model is perfect.
D) The model is random.

Answer: B

---

19. What is LabelBinarizer used for?

A) Scaling features
B) Encoding multi-class labels
C) Splitting data
D) Cleaning missing values

Answer: B

---

20. What does macro averaging do?

A) Treats all samples equally
B) Treats all classes equally
C) Treats all features equally
D) Ignores classes

Answer: B

---

21. What does micro averaging do?

A) Treats all classes equally
B) Treats all predictions equally
C) Treats only minority class equally
D) Ignores sample count

Answer: B

---

22. Decision trees used for classification are what kind of learning method?

A) Unsupervised
B) Supervised
C) Reinforcement
D) Self-organizing only

Answer: B

---

23. What does the .fit() method do?

A) Predicts outputs
B) Trains the model
C) Splits the data
D) Evaluates the result

Answer: B

---

24. What is the test set used for?

A) Training the model
B) Evaluating performance
C) Encoding features
D) Balancing classes

Answer: B

---

25. Which four terms appear in a confusion matrix?

A) TP
B) FP
C) TN
D) FN

Answer: A, B, C, D

---

26. What does high specificity usually reduce?

A) True positives
B) False positives
C) False negatives
D) True negatives

Answer: B

---

27. What does high recall usually reduce?

A) False positives
B) False negatives
C) True negatives
D) Total observations

Answer: B

---

28. What is the root node in a decision tree?

A) The last node
B) The first split or starting point
C) The leaf node only
D) The output label

Answer: B

---

29. What is a leaf node in a decision tree?

A) A split node
B) A final prediction node
C) The root node
D) An input variable

Answer: B

---

30. What usually happens when min_samples_leaf is set high?

A) The tree becomes more complex.
B) The tree becomes simpler.
C) The model stops training.
D) The tree always becomes deeper.

Answer: B

---

31. What is Gini impurity mainly used for?

A) Encoding text
B) Choosing splits
C) Scaling values
D) Plotting curves

Answer: B

---

32. Decision trees can be used for which tasks?

A) Regression only
B) Classification only
C) Both classification and regression
D) Neither

Answer: C

---

33. What is the AUC of a perfect model?

A) 0
B) 0.5
C) 1
D) 2

Answer: C

---

34. When does data leakage occur?

A) When test data influences training
B) When the data is split correctly
C) When categories are encoded
D) When features are scaled

Answer: A

---

35. What does a feature mean in machine learning?

A) Output variable
B) Input variable
C) Error term
D) Label encoding

Answer: B

---

36. What does a target variable mean?

A) Input variable
B) Output variable
C) Noise only
D) Feature group only

Answer: B

---

37. What does an imbalanced dataset mean?

A) Equal-sized classes
B) Unequal-sized classes
C) Clean data only
D) Purely numeric data

Answer: B

---

38. What does Random Forest usually improve?

A) Speed only
B) Stability
C) File size only
D) Number of labels only

Answer: B

---

39. What does cross-validation provide?

A) Faster training only
B) A better performance estimate
C) Less data
D) Random output

Answer: B

---

40. If precision is high, what is usually true?

A) There are few false positives.
B) There are few false negatives.
C) There are few true negatives.
D) There are few true positives.

Answer: A

---

41. If recall is high, what is usually true?

A) There are few false positives.
B) There are few false negatives.
C) There are few true negatives.
D) There are few total samples.

Answer: B

---

42. Which point is ideal on a ROC curve?

A) Bottom-left
B) Top-left
C) Top-right
D) Center

Answer: B

---

43. Which point is ideal on a PR curve?

A) Bottom-left
B) Top-right
C) Center
D) Bottom-right

Answer: B

---

44. Decision tree splits are mainly based on what?

A) Randomness only
B) Impurity reduction
C) File size
D) Time order only

Answer: B

---

45. On which data should final model evaluation be based?

A) Training data
B) Test data
C) Random data only
D) No data

Answer: B

---

46. What usually must happen to categorical data before using CART in scikit-learn?

A) It must always be removed.
B) It must be converted to numeric form.
C) It must be ignored.
D) It must be randomized.

Answer: B

---

47. What does SMOTE create?

A) Random noise
B) Synthetic minority samples
C) New labels only
D) New splits only

Answer: B

---

48. What usually happens when tree depth increases a lot?

A) Overfitting risk increases
B) Underfitting increases
C) Nothing changes
D) Accuracy always improves on test data

Answer: A

---

49. What does pruning mainly reduce?

A) Data volume
B) Model complexity
C) Number of labels
D) Number of observations

Answer: B

---

50. What does the best model choice depend on most strongly?

A) Accuracy only
B) Business cost and context
C) Data size only
D) Number of features only

Answer: B

Online Assessment Questions

1. What is a primary limitation of deep feedforward networks?

A) Overfitting only
B) Vanishing gradient
C) Slow training only
D) No output layer

Answer: B

---

2. CNNs are best suited for which type of data?

A) Text only
B) Images
C) Time series only
D) Graphs only

Answer: B

---

3. What is the key idea behind transformers?

A) Convolution only
B) Recurrence only
C) Self-attention
D) Clustering

Answer: C

---

4. What does positional encoding mainly help with?

A) Speed only
B) Preserving the order of words
C) Accuracy only
D) Loss calculation only

Answer: B

---

5. What does fine-tuning mean?

A) Training a model from scratch
B) Adapting a pretrained model
C) Removing layers only
D) Testing only

Answer: B

---

6. What does the hidden state in an RNN do?

A) It resets every time without memory.
B) It stores memory across sequence steps.
C) It deletes information.
D) It has no purpose.

Answer: B

---

7. In which setting is an encoder-decoder architecture commonly used?

A) CNN-only image tasks
B) Sequence-to-sequence tasks
C) Clustering only
D) Regression only

Answer: B

---

8. What advantage do RNNs have over basic feedforward networks for sequences?

A) Lower memory use only
B) Sequence memory
C) Smaller file size only
D) No weights

Answer: B

---

9. What do shared weights in an RNN mean?

A) Different weights at every step
B) The same weights are reused across steps
C) Random weights are used
D) No weights are used

Answer: B

---

10. What does the output gate in an LSTM control?

A) Input value only
B) Exposure of memory to the output
C) Forgetting only
D) Training speed

Answer: B

Week 9

Concepts

1. Risk Management Process

Risk management is the process of identifying risks, assessing them, and then controlling or reducing them. It is a structured way to protect systems from failure, harm, and loss.

Risk identification means finding possible threats such as data leakage, model failure, bias, or security attacks.

Risk assessment means estimating how likely a threat is and how serious the impact would be.

Risk control or mitigation means taking actions to reduce the probability or impact of those risks.

2. Quantitative Risk Assessment

Quantitative risk assessment uses numbers rather than only qualitative judgment.

A common framework treats risk as a function of threat probability, success probability of attack, exposure of assets, and asset value.

Measurement error is the uncertainty or inaccuracy in estimating these values.

3. Levels of Risk Assessment

Risk can be assessed at the individual level, the organizational level, or the national level. The same technical failure may have very different consequences at each level.

A measurable impact of AI system failure is often financial cost, such as the cost created by a wrong automated decision.

4. Threat Categories and ML Attacks

Threats to information systems can be broadly classified as malicious or unintentional.

Malicious threats are intentional attacks. Unintentional threats come from accidents, mistakes, bugs, or poor design.

Data poisoning is a malicious attack where the training data is corrupted so the model learns wrong patterns.

Adversarial input attacks slightly modify inputs to cause wrong predictions, even though the changes may be hard for humans to notice.

5. Environmental Concerns and Bias

Large AI models consume a great deal of energy during training and deployment. This creates environmental concern because of high electricity use and carbon emissions.

A common criticism of large-scale AI systems is that the training data may not represent the diversity of the real population. This can lead to bias and unfair outputs.

Example: If a face recognition model is trained mostly on one group of faces, it may perform worse on underrepresented groups. This becomes a fairness and risk management problem at the same time.

MCQs

1. Which sequence best represents the typical risk management process?

A) Risk control, risk identification, risk assessment
B) Risk identification, risk assessment, risk control
C) Risk assessment, risk mitigation, risk identification
D) Risk monitoring, risk identification, risk response

Answer: B

---

2. What does risk identification mean?

A) Reducing risks
B) Finding risks
C) Measuring revenue
D) Ignoring threats

Answer: B

---

3. What does risk assessment mainly focus on?

A) Detection only
B) Probability and impact
C) Coding style
D) Hardware brand

Answer: B

---

4. What does quantitative risk assessment mainly use?

A) Opinions only
B) Numbers and measurable values
C) Guessing only
D) Rules without data

Answer: B

---

5. Which factors commonly appear in a quantitative risk model?

A) Only cost
B) Only probability
C) Threat probability, success probability, exposure, and asset value
D) Time only

Answer: C

---

6. What does measurement error mean in risk modeling?

A) A coding bug only
B) A wrong screen output only
C) Uncertainty in estimated values
D) A hardware failure only

Answer: C

---

7. Which levels are commonly used for AI risk assessment?

A) Technical only
B) Individual, organizational, and national
C) Financial only
D) Social only

Answer: B

---

8. Which of the following is a measurable impact of AI failure?

A) Loss of trust only
B) Psychological stress only
C) Financial cost
D) Reputation only

Answer: C

---

9. Threats are often classified into which two categories?

A) Internal and external
B) Malicious and unintentional
C) Logical and physical
D) Manual and automatic

Answer: B

---

10. What is a malicious threat?

A) An accident only
B) An intentional attack
C) A random error only
D) A natural issue only

Answer: B

---

11. What is an unintentional threat?

A) A planned attack
B) Human error or accident
C) Malware only
D) A hack only

Answer: B

---

12. Which of the following is an example of a malicious ML attack?

A) Overfitting
B) Data poisoning
C) Low accuracy
D) Normal noise

Answer: B

---

13. During which phase does data poisoning mainly happen?

A) During testing only
B) During training
C) During deployment only
D) During storage only

Answer: B

---

14. What is the main goal of an adversarial attack?

A) To improve accuracy
B) To reduce cost
C) To cause a wrong prediction
D) To increase speed

Answer: C

---

15. How does an adversarial input attack usually work?

A) By changing labels only
B) By modifying inputs to fool the model
C) By removing all data
D) By adding features to the database only

Answer: B

---

16. What is a major environmental concern related to large AI systems?

A) Memory usage only
B) Energy consumption and carbon emissions
C) Storage location only
D) Display quality only

Answer: B

---

17. When does AI bias commonly occur?

A) When data is large
B) When data lacks diversity
C) When the model is small
D) When hardware fails

Answer: B

---

18. What can AI bias lead to?

A) Faster models
B) Fair decisions
C) Unfair outcomes
D) Lower cost only

Answer: C

---

19. What does risk control mean?

A) Ignoring the risk
B) Reducing or managing the risk
C) Increasing the risk
D) Measuring risk only

Answer: B

---

20. In a risk model, what does exposure refer to?

A) Attack type only
B) The degree to which an asset is vulnerable or exposed
C) Cost only
D) Speed only

Answer: B

---

21. What does asset value mean in risk assessment?

A) The coding language used
B) The importance or worth of the asset
C) The screen size only
D) The model type only

Answer: B

---

22. What does success probability mean in a risk formula?

A) Chance that the attack works
B) Chance that the company grows
C) Cost of the attack
D) Time required for training

Answer: A

---

23. What does threat probability mean?

A) Damage amount only
B) Chance that the threat occurs
C) Cost only
D) Model accuracy only

Answer: B

---

24. What may incorrect estimation of risk parameters create?

A) Better protection automatically
B) Measurement error
C) Faster systems
D) Less cost automatically

Answer: B

---

25. Organizational risk mainly affects whom or what?

A) One individual only
B) The company or organization
C) The country only
D) A single file only

Answer: B

---

26. National risk mainly affects what?

A) One company only
B) One individual only
C) A country or nation
D) One model only

Answer: C

---

27. Which of the following is an indirect impact of AI failure?

A) Financial cost
B) Revenue loss only
C) Loss of trust
D) Direct compensation cost

Answer: C

---

28. Which of the following is a direct impact of AI failure?

A) Psychological stress only
B) Reputational damage only
C) Financial loss
D) Trust decline only

Answer: C

---

29. Which of the following is an example of a malicious attack?

A) A typo
B) Data poisoning
C) An accidental crash
D) A bug caused by oversight

Answer: B

---

30. Which of the following is an example of an unintentional threat?

A) Hacking
B) A coding error
C) Malware injection
D) An adversarial attack

Answer: B

---

31. What is an adversarial example?

A) A clean input
B) A slightly modified input that fools the model
C) A random dataset
D) A normal label only

Answer: B

---

32. High AI energy use contributes to what?

A) Faster training only
B) Carbon emissions
C) Less data use
D) Higher fairness automatically

Answer: B

---

33. AI bias mainly comes from what source?

A) Hardware only
B) Training data
C) CPU brand only
D) GPU memory only

Answer: B

---

34. Which of the following is an example of risk mitigation?

A) Ignoring threats
B) Applying prevention and protection measures
C) Removing the entire system immediately
D) Deleting all data without analysis

Answer: B

---

35. Which of the following is not part of the risk management process?

A) Identification
B) Assessment
C) Control
D) Coding style selection

Answer: D

---

36. What should risk assessment rely on as much as possible?

A) Luck
B) Data and evidence
C) Random choice
D) Guesswork only

Answer: B

---

37. What usually happens when exposure increases?

A) Safety increases
B) Risk increases
C) Accuracy increases automatically
D) Speed increases automatically

Answer: B

---

38. What does high asset value usually imply?

A) Lower impact
B) Higher possible damage
C) No risk
D) Smaller losses

Answer: B

---

39. Adversarial attacks are best described as what?

A) Random accidents
B) Intentional attacks
C) Natural failures
D) Hardware issues only

Answer: B

---

40. What does data poisoning mainly target?

A) Output display only
B) Training data
C) Hardware only
D) File storage only

Answer: B

---

41. When is risk typically highest?

A) Low probability and low impact
B) Low value and low exposure
C) High probability and high impact
D) Low exposure only

Answer: C

---

42. Which of the following is part of a common risk formula?

A) CPU type
B) Exposure
C) RAM size
D) GPU brand

Answer: B

---

43. What does bias mainly affect in AI systems?

A) Hardware
B) Fairness
C) Screen brightness
D) Storage speed

Answer: B

---

44. Why is AI failure cost considered measurable?

A) Because it is always emotional
B) Because it can be expressed financially
C) Because it is always hidden
D) Because it has no business impact

Answer: B

---

45. What does the term threat mean?

A) Protection
B) Danger or possible harm
C) Safety feature
D) Optional add-on

Answer: B

---

46. Which of the following is an unintentional threat?

A) Malware injection
B) Human error
C) Coordinated attack
D) Hacking attempt

Answer: B

---

47. Which of the following is a malicious threat?

A) Mistake
B) Intentional attack
C) Software typo only
D) Accidental crash

Answer: B

---

48. What does risk control try to reduce?

A) Accuracy
B) Risk
C) Data size
D) Feature count

Answer: B

---

49. What kind of system can AI bias create?

A) A fair system
B) An unfair system
C) A faster system only
D) A smaller system only

Answer: B

---

50. What is the main goal of risk management?

A) To increase risk
B) To ignore risk
C) To reduce risk
D) To hide risk

Answer: C

Online Assessment Questions

1. Which of the following best represents the typical process of risk management?

A) Risk control, risk identification, risk assessment
B) Risk identification, risk assessment, risk control
C) Risk assessment, risk mitigation, risk identification
D) Risk monitoring, risk identification, risk response

Answer: B

2. A quantitative risk assessment framework models risk as a function of which factors?

A) Organizational culture, governance strength, and technology maturity
B) Regulatory pressure, innovation capability, and investment cost
C) Threat probability, success probability of attack, exposure of assets, and asset value
D) Data volume, model complexity, and infrastructure capacity

Answer: C

3. In risk measurement models, what does measurement error refer to?

A) The probability that a threat will occur
B) The uncertainty or inaccuracy in estimating risk parameters
C) The financial loss due to an attack
D) The technical failure of a risk detection system

Answer: B

4. At which levels can AI-enabled risk assessments typically be conducted?

A) Technical, financial, and operational
B) Individual, organizational, and national
C) Algorithmic, computational, and infrastructural
D) Ethical, regulatory, and cultural

Answer: B

5. Which of the following is a tangible and measurable impact of AI system failure?

A) Loss of trust in technology
B) Psychological distress caused by system errors
C) Cost resulting from incorrect automated decisions
D) Long-term reputational damage

Answer: C

6. Threats to information systems can broadly be classified into which two categories?

A) Strategic and operational
B) Internal and external
C) Malicious and unintentional
D) Physical and digital

Answer: C

---

7. Which of the following is an example of a malicious attack on machine learning systems?

A) Model overfitting during training
B) Data poisoning during the training phase
C) Poor feature selection in model design
D) Insufficient computing resources

Answer: B

---

8. What does an adversarial input attack primarily aim to do?

A) Increase computational efficiency
B) Improve model generalization
C) Manipulate inputs to cause incorrect predictions
D) Reduce model bias during training

Answer: C

---

9. Why do large AI models raise environmental concerns?

A) High licensing costs only
B) Extensive energy consumption and carbon emissions during training
C) Lack of regulation only
D) Limited training data only

Answer: B

---

10. What is a common criticism of large AI models trained on internet-scale data?

A) They rely only on proprietary datasets.
B) Their training data may not represent the diversity of global populations.
C) They cannot process structured data.
D) They cannot learn language patterns.

Answer: B

Week 10

Concepts

1. Clustering and Store Segmentation

Clustering is an unsupervised learning method that groups similar observations together without pre-labeled classes.

Store segmentation applies clustering to stores instead of customers so that similar stores can be managed using similar strategies.

Cross-sectional data is data collected at one point in time across many units. This is important because clustering should not confuse time-based change with true similarity.

Cluster profiling means interpreting the meaning of each cluster by examining average values and business characteristics.

2. K-Means and Data Preparation

K-Means partitions observations into K clusters based on distance from centroids. It works best when features are scaled and K is chosen carefully.

Multicollinearity means variables are highly correlated and duplicate information. Redundant variables can distort clustering.

Scaling using a method such as StandardScaler is important because K-Means is distance-based, and large numeric values can dominate the result.

Elbow method plots SSE against K. The bend or elbow point suggests a good number of clusters.

SSE, or Sum of Squared Errors, measures how tight the clusters are. Lower SSE generally means better clustering.

3. RFM Analysis and Customer Value

RFM analysis segments customers using Recency, Frequency, and Monetary value.

Recency means how recently the customer purchased. Lower time since the last purchase is better and often the strongest indicator of current engagement.

Frequency measures how often the customer purchases. High frequency often signals loyalty.

Monetary measures how much the customer spent. It indicates revenue contribution.

CLV is customer lifetime value, the predicted future value of a customer. RFM helps estimate it.

4. Customer Segment Meanings

True Friends have high recency quality, high frequency, and high monetary value. They are strong loyal customers.

Butterflies spend a lot but are not very frequent. The firm should maximize short-term profit from them.

Barnacles buy often but spend little. The strategy is to increase their average spend.

Strangers have low values across RFM and often do not deserve major investment.

5. Data Cleaning and Aggregation

Data cleaning removes errors such as impossible or negative values that should not exist in standard purchases.

Returns are treated as negative transactions so they reduce total revenue.

Pivot tables are used to aggregate RFM values: SUM for monetary, COUNT for frequency, and MAX date for recency calculations.

Scaling can also convert raw RFM values into a common score range such as 1 to 10. Reverse scaling is often used for recency because lower days since purchase should produce a higher score.

NTILE(n) divides data into equal-sized groups such as quintiles or deciles.

fit_predict() trains a model and returns cluster labels in one step.

3D visualization is useful in RFM because there are three main features to display.

6. LLM and Voice Assistant Topics in Online Assessment

This week?s online assessment also touches separate AI application topics such as LLMs, RAG, prompt injection, vector similarity, voice assistants, and agentic workflows. These are included in the assessment section exactly as provided.

> *Example: If a customer bought very recently, buys often, and spends a lot, the customer may be tagged as a True Friend. If another customer spends a lot once in a while, that customer fits better in the Butterfly segment.*

MCQs

1. What is clustering?

A) Supervised learning
B) Grouping similar data points
C) Regression technique
D) Classification only

Answer: B

---

2. Clustering belongs to which type of learning?

A) Supervised learning
B) Unsupervised learning
C) Semi-supervised learning
D) Reinforcement learning

Answer: B

---

3. Store segmentation focuses on grouping which entities?

A) Customers
B) Products
C) Stores
D) Employees

Answer: C

---

4. What is cross-sectional data?

A) Data collected over many time periods for one case
B) Data collected at one point in time across many units
C) Random simulated data
D) Forecasted data only

Answer: B

---

5. Why might analysts reduce a long history into a short common window before clustering?

A) To improve keyboard speed
B) To reduce storage only
C) To remove time variance from the comparison
D) To avoid plotting

Answer: C

---

6. What does cluster profiling mainly help with?

A) Reducing data size
B) Interpreting clusters in business terms
C) Scaling data only
D) Training labels

Answer: B

---

7. K-Means mainly uses what to assign clusters?

A) Probability only
B) Distance from centroids
C) Decision rules only
D) Linguistic features

Answer: B

---

8. What does multicollinearity mean?

A) Missing data
B) Highly correlated variables
C) Noisy labels only
D) Clean data only

Answer: B

---

9. Why is scaling important in K-Means?

A) Because the dataset is large
B) Because K-Means is distance-based
C) Because it removes labels
D) Because it stores data better

Answer: B

---

10. What does the elbow method help find?

A) The best model language
B) The optimal K value
C) The best feature name
D) The best dataset source

Answer: B

---

11. What does SSE measure in clustering?

A) Accuracy
B) Cluster tightness
C) Speed only
D) Storage only

Answer: B

---

12. What does lower SSE usually indicate?

A) Worse clusters
B) Better clusters
C) No change
D) Random assignment

Answer: B

---

13. What does recency measure in RFM analysis?

A) Spending amount
B) Time since last purchase
C) Purchase frequency only
D) Profit margin

Answer: B

---

14. What does frequency measure in RFM analysis?

A) Time only
B) Number of purchases
C) Profit only
D) Loss only

Answer: B

---

15. What does monetary measure in RFM analysis?

A) Time
B) Count
C) Spending
D) Profit margin only

Answer: C

---

16. Which RFM factor is often the strongest predictor of engagement?

A) Monetary
B) Frequency
C) Recency
D) CLV

Answer: C

---

17. Who are True Friends in RFM segmentation?

A) Customers with low RFM values
B) Customers with high RFM values
C) Customers with only high monetary value
D) Customers with only high frequency

Answer: B

---

18. Who are Butterflies?

A) Highly loyal low spenders
B) High spenders with low frequency
C) Low spenders only
D) Customers with no activity

Answer: B

---

19. Who are Barnacles?

A) High spenders only
B) Low loyalty premium buyers
C) Frequent but low-spend customers
D) New users only

Answer: C

---

20. Who are Strangers?

A) High-value customers
B) Medium-value customers
C) Low-value customers across RFM
D) Loyal premium customers

Answer: C

---

21. What does data cleaning mainly remove?

A) Useful features
B) Errors and invalid values
C) Models
D) Labels

Answer: B

---

22. In ordinary sales data, what may a negative price indicate?

A) Profit
B) An error or return effect
C) Tax only
D) A new label

Answer: B

---

23. How are returns usually treated in transactional analysis?

A) Removed completely
B) Counted as positive sales
C) Recorded as negative values
D) Ignored permanently

Answer: C

---

24. In a pivot table, what does SUM usually give for RFM?

A) Frequency
B) Recency
C) Monetary value
D) Average order gap

Answer: C

---

25. In a pivot table, what does COUNT usually give for RFM?

A) Recency
B) Frequency
C) Monetary value
D) Average value

Answer: B

---

26. In a pivot table, what does MAX date usually help identify?

A) First purchase
B) Last purchase date
C) Frequency only
D) Spending only

Answer: B

---

27. What is the main purpose of scaling scores in RFM?

A) To create a larger range
B) To place values on a common scale
C) To randomize records
D) To create binary variables

Answer: B

---

28. Why is reverse scaling used for recency?

A) Because frequency is negative
B) Because lower recency should get a better score
C) Because monetary must be removed
D) Because CLV is always negative

Answer: B

---

29. What does NTILE(n) do?

A) Creates random groups
B) Divides data into equal groups
C) Creates unequal groups
D) Deletes outliers

Answer: B

---

30. What does fit_predict() do?

A) Only trains
B) Only predicts
C) Trains and predicts cluster labels
D) Does nothing

Answer: C

---

31. Why is a 3D plot useful in RFM analysis?

A) Because there are 2 features
B) Because there are 3 main features
C) Because there are 4 dimensions
D) Because it always improves accuracy

Answer: B

---

32. What usually happens to SSE as K increases?

A) It increases
B) It decreases
C) It stays fixed
D) It becomes random

Answer: B

---

33. What can happen if K is set too large?

A) Underfitting only
B) Over-segmentation or overfitting-like fragmentation
C) No clusters form
D) The data disappears

Answer: B

---

34. Does K-Means require labeled target data?

A) Yes
B) No
C) Only for classification
D) Only for regression

Answer: B

---

35. What is a cluster center in K-Means?

A) Median only
B) Mean point or centroid
C) Mode only
D) Sum of values

Answer: B

---

36. Which distance is commonly used in K-Means?

A) Manhattan distance
B) Euclidean distance
C) Hamming distance
D) Cosine distance only

Answer: B

---

37. What does CLV predict?

A) Past value only
B) Future customer value
C) Present stock only
D) None

Answer: B

---

38. RFM analysis is mainly based on what kind of data?

A) Demographics only
B) Transactions
C) Weather data
D) Surveys only

Answer: B

---

39. What does high frequency usually signal?

A) Low loyalty
B) High loyalty
C) Low spending automatically
D) No meaningful trend

Answer: B

---

40. What does low recency usually mean?

A) The customer bought recently
B) The customer bought long ago
C) The customer never bought
D) The customer spent a lot

Answer: A

---

41. What does feature reduction help with before clustering?

A) Increasing redundancy
B) Removing duplicate information
C) Adding noise
D) Slowing the algorithm

Answer: B

---

42. Which method is associated with grouping similar stores in this week?

A) Regression
B) Clustering
C) Classification only
D) NLP

Answer: B

---

43. In store segmentation, what is the unit of analysis?

A) Customer
B) Store
C) Product
D) Time period only

Answer: B

---

44. What does cluster profiling give to a cluster?

A) Accuracy only
B) Business meaning
C) Speed only
D) Error only

Answer: B

---

45. What does scaling prevent in distance-based methods?

A) All errors
B) Dominance by large-scale variables
C) All loss
D) All noise

Answer: B

---

46. What is usually the best K value?

A) The maximum possible K
B) The minimum possible K
C) A value that is both well-supported and interpretable
D) A random value

Answer: C

---

47. What is a good strategy for Butterfly customers?

A) Long-term retention at any cost
B) Ignore them fully
C) Maximize current profit opportunities
D) Reduce all contact

Answer: C

---

48. What is a good strategy for Barnacle customers?

A) Ignore them
B) Increase their spending per purchase
C) Remove them immediately
D) Replace them with strangers

Answer: B

---

49. What is a common strategy for Stranger customers?

A) Invest heavily in them
B) Largely ignore or spend little on them
C) Reward them as loyal customers
D) Treat them as premium customers

Answer: B

---

50. What is a common strategy for True Friends?

A) Ignore them
B) Retain and nurture them
C) Reduce service quality
D) Replace them

Answer: B

Online Assessment Questions

1. At its most fundamental architectural level, what is the primary operational objective of a base Large Language Model?

A) To understand and replicate human emotional intent
B) To perform recursive semantic analysis on a global knowledge base
C) To predict the most statistically likely subsequent token in a sequence
D) To execute logic-based reasoning through symbolic AI integration

Answer: C

---

2. Which parameter directly influences the probability distribution of the next token by flattening or sharpening the logits before softmax is applied?

A) Top-P
B) Top-K
C) Temperature
D) Frequency penalty

Answer: C

---

3. In the context of LLM security, what constitutes a prompt injection attack?

A) Using brute force to guess an API key
B) Overwhelming the context window with repetitive tokens
C) Crafting inputs that make the model ignore system instructions and follow adversarial user commands
D) Injecting malicious Python code into a vector database

Answer: C

4. How does Retrieval-Augmented Generation primarily mitigate the issue of knowledge cutoff?

A) By dynamically fine-tuning base weights on real-time data
B) By expanding the model?s internal parameter count
C) By fetching relevant document chunks from an external database and appending them to the prompt context
D) By increasing the model temperature

Answer: C

5. Which metric is most commonly used in vector databases to calculate semantic distance between embeddings?

A) Euclidean distance
B) Cosine similarity
C) Hamming distance
D) Pearson correlation coefficient

Answer: B

6. What is the environmental significance of prompt efficiency in large-scale enterprise deployments?

A) Shorter prompts require less storage
B) Minimizing token usage reduces electrical energy consumed by GPUs during inference
C) It eliminates cooling systems
D) It prevents data corruption

Answer: B

7. In which ways does an agentic workflow differ from a linear RAG pipeline?

A) It can dynamically select tools based on input
B) It can involve self-correction loops
C) It requires zero token processing
D) It can branch into different logic paths

Answer: A, B, D

8. Which of the following best describes the Dolphin Attack vulnerability in AI voice systems?

A) Brute-force wake word attack
B) Use of ultrasonic sound waves inaudible to humans to deliver malicious commands
C) Social engineering via voice mimicry
D) Exploiting similar app names

Answer: B

9. From an MLOps perspective, why is user voice data often stored instead of deleted immediately?

A) Legal compliance only
B) To reduce hardware usage
C) To provide labeled data for supervised training and improve accuracy
D) To back up communication logs only

Answer: C

10. What primary technical mechanism allows voice assistants like Alexa to respond without a physical interface?

A) Continuous cloud transcription only
B) Localized wake-word detection hardware
C) Biometric identification only
D) Predictive intent modeling only

Answer: B

Week 11

Concepts

1. Fairness, Diversity, and Inclusion

Diversity means having people from different backgrounds, identities, and communities represented in a group.

Inclusion means making sure those people are not only present but also valued, heard, and able to participate meaningfully.

Systematic relative disadvantage means some groups repeatedly face unfair treatment because of bias built into social systems.

Equality of opportunity in the middle-view sense recognizes past disadvantage and tries to create fair chances rather than pretending all groups started from the same position.

2. Responsibility and Accountability in AI

The responsibility gap appears because AI systems can make impactful decisions without having consciousness or moral agency, while humans may not fully control every outcome.

Distributed responsibility means responsibility is shared across developers, firms, deployers, users, regulators, and other stakeholders.

3. AI Risk Management Responses

Defense means using safeguards to prevent or block harmful behavior.

Mitigation means reducing the impact when harm cannot be fully prevented.

Transfer means shifting risk to another party, such as through insurance, outsourcing, or contractual arrangements.

Acceptance means deciding to tolerate a risk when it is low, unavoidable, or not worth the cost of prevention.

4. AI Evaluation Measures

Language Modeling Score (LMS) measures how often a model chooses a sensible or logical answer instead of a nonsensical one.

Stereotype Score (SS) measures how often a model chooses stereotyped responses. A higher stereotype score implies more bias.

5. ANN Concepts

ANN training adjusts weights to reduce prediction error.

Backpropagation sends error backward through the network so the weights can be updated.

A feedforward network moves information from input to output without loops. Hidden layers help the model learn more complex patterns.

The bias-variance tradeoff still matters: a model that is too simple underfits, while a model that is too complex overfits.

6. Time Series and Financial Signals

Time series data is collected across time intervals, such as daily stock prices or hourly demand.

Technical analysis uses past prices and trading patterns to estimate future movement, while fundamental analysis uses broader economic and company information.

RSI is a momentum indicator. A very high RSI may signal overbought conditions, while a very low RSI may signal oversold conditions.

7. Implementation Concepts

Data partitioning in time series should use older data for training and newer data for testing to avoid look-ahead bias.

Feature engineering creates useful variables such as SMA, RSI, and momentum.

Data synchronization ensures that all time-based features line up correctly on the same timeline before modeling.

> *Example: If a trading model uses tomorrow?s price while pretending to predict today?s movement, it is cheating. Proper time-series partitioning prevents this kind of look-ahead bias.*

MCQs

1. What does diversity mainly refer to?

A) Equal pay only
B) Representation of different groups
C) Bias removal only
D) Legal rights only

Answer: B

---

2. What does inclusion mainly ensure?

A) Hiring only
B) Participation and belonging
C) Equal salary only
D) Legal equality only

Answer: B

---

3. What does systematic relative disadvantage mean?

A) One-time unfairness
B) Repeated unfair treatment of some groups
C) Equal opportunity automatically
D) Legal protection only

Answer: B

---

4. What does the middle view of equality of opportunity recognize?

A) It ignores the past.
B) It adjusts for past disadvantage.
C) It demands identical outcomes.
D) It uses random selection.

Answer: B

---

5. Why does a responsibility gap arise in AI?

A) Because AI is perfect
B) Because AI lacks consciousness and humans do not fully control every outcome
C) Because humans are absent
D) Because data is always missing

Answer: B

---

6. What does distributed responsibility mean?

A) One person is fully responsible
B) Responsibility is shared across actors
C) No one is responsible
D) Only the AI is responsible

Answer: B

---

7. What does defense mean in AI risk management?

A) Ignore the risk
B) Prevent or block the risk
C) Transfer the risk
D) Accept the risk only

Answer: B

---

8. What does mitigation mean in AI risk management?

A) Remove all risk completely
B) Reduce the impact of risk
C) Ignore risk
D) Increase risk

Answer: B

---

9. What does transfer mean in AI risk management?

A) Reduce risk only
B) Shift risk to another party
C) Eliminate all risk
D) Ignore the risk

Answer: B

---

10. What does acceptance mean in AI risk management?

A) Prevent the risk
B) Accept the risk as it is
C) Transfer the risk
D) Reduce the risk only

Answer: B

---

11. What does LMS measure?

A) Bias only
B) Logic versus nonsense in model outputs
C) Speed only
D) Memory only

Answer: B

---

12. What does Stereotype Score measure?

A) Accuracy only
B) Bias level
C) Speed only
D) Memory only

Answer: B

---

13. What does a high stereotype score usually indicate?

A) Less bias
B) More bias
C) Better logic
D) Faster model

Answer: B

---

14. What is the main aim of ANN training?

A) Increasing data size
B) Reducing prediction error
C) Removing layers
D) Increasing speed only

Answer: B

---

15. What does backpropagation do?

A) Sends data only forward
B) Sends error backward for weight updates
C) Cleans data
D) Makes final predictions only

Answer: B

---

16. What is a feedforward network?

A) A network with loops
B) A network with one-directional flow
C) A network with random flow
D) A network with no input

Answer: B

---

17. What do hidden layers mainly help with?

A) Storage only
B) Learning complex patterns
C) Output labeling only
D) Input capture only

Answer: B

---

18. What does overfitting mean?

A) Poor training effort only
B) Memorizing training data too closely
C) No learning
D) Random output only

Answer: B

---

19. What does underfitting mean?

A) The model is too simple.
B) The model is too complex.
C) The model is perfect.
D) The model is random.

Answer: A

---

20. What is time series data?

A) Random observations only
B) Data ordered across time
C) Static grouped data only
D) Cluster labels only

Answer: B

---

21. What is panel data?

A) Data from one entity only
B) Multiple entities observed over time
C) Static one-time data only
D) Random data

Answer: B

---

22. What does technical analysis mainly use?

A) Weather data
B) Price and market data
C) Cultural data only
D) Legal data only

Answer: B

---

23. What does fundamental analysis mainly use?

A) Price charts only
B) Economic and business information
C) Random noise
D) Graphics only

Answer: B

---

24. What does an RSI above 80 often suggest?

A) Strong buy only
B) Overbought condition
C) No movement
D) Data error

Answer: B

---

25. What does an RSI below 20 often suggest?

A) Strong sell only
B) Oversold condition
C) No trading
D) Feature mismatch

Answer: B

---

26. What does momentum measure?

A) Speed of change
B) Price level only
C) Time only
D) Volume only

Answer: A

---

27. What does data partitioning mainly prevent in time series work?

A) Overfitting only
B) Look-ahead bias
C) Noise only
D) Fairness bias only

Answer: B

---

28. What does feature engineering mean?

A) Deleting all raw data
B) Creating useful new input variables
C) Storing data only
D) Visualizing charts only

Answer: B

---

29. What does synchronization ensure in time-series modeling?

A) Same time alignment across features
B) Faster training only
C) More layers
D) Less data always

Answer: A

---

30. What are hyperparameters?

A) Learned weights only
B) Predefined settings chosen before training
C) Outputs only
D) Inputs only

Answer: B

---

31. What is the ReLU activation function?

A) Always negative
B) max(0, x)
C) Always linear with no threshold
D) Random

Answer: B

---

32. What is a perceptron?

A) A dataset
B) An artificial neuron
C) An output file
D) A plotting function

Answer: B

---

33. What does gradient descent do?

A) Increase error
B) Minimize error
C) Search randomly only
D) Split the data

Answer: B

---

34. What does it mean when a model is called a black box?

A) It is fully transparent.
B) Its internal reasoning is hard to interpret.
C) It is always simple.
D) It is always linear.

Answer: B

---

35. What does one epoch mean in ANN training?

A) One data point
B) One full pass through the dataset
C) One neuron only
D) One layer only

Answer: B

---

36. What can too many hidden layers increase if not controlled well?

A) Underfitting only
B) Overfitting risk
C) Data quality automatically
D) Fairness automatically

Answer: B

---

37. Which tradeoff balances model simplicity and flexibility?

A) Precision-recall tradeoff
B) Bias-variance tradeoff
C) Cost-benefit tradeoff
D) Input-output tradeoff

Answer: B

---

38. What does defense mainly try to do?

A) Shift risk to another party
B) Prevent risk before it causes harm
C) Accept every risk
D) Ignore all threats

Answer: B

---

39. What does mitigation mainly try to do?

A) Remove all risk always
B) Reduce impact when harm occurs
C) Transfer cost only
D) Eliminate data

Answer: B

---

40. What does transfer mean in risk handling?

A) Blocking every risk
B) Shifting the burden to another party
C) Accepting all harm
D) Creating more risk

Answer: B

---

41. What does acceptance mean in risk handling?

A) Preventing risk fully
B) Tolerating the risk without additional action
C) Outsourcing the model always
D) Deleting the system

Answer: B

---

42. Why is inclusion different from diversity?

A) Inclusion focuses on participation, not just representation.
B) Inclusion focuses only on salary.
C) Diversity and inclusion mean exactly the same thing.
D) Inclusion applies only to algorithms.

Answer: A

---

43. What does systematic disadvantage suggest?

A) A random one-time event
B) A repeated structural problem
C) Perfect equality
D) No social pattern

Answer: B

---

44. What kind of score becomes problematic when it is high in stereotype evaluation?

A) LMS
B) SS
C) Accuracy only
D) Precision only

Answer: B

---

45. What does a high LMS suggest?

A) The model often chooses logical answers.
B) The model is highly biased.
C) The model is always unfair.
D) The model cannot reason at all.

Answer: A

---

46. Why should newer data usually be kept for testing in time series?

A) To avoid future information leaking into training
B) To make training easier only
C) To balance labels
D) To remove all bias

Answer: A

---

47. Which of the following is an example of a feature-engineered indicator?

A) SMA
B) Raw company name only
C) Page color only
D) File extension only

Answer: A

---

48. What does feedforward mean?

A) Data moves backward first
B) Data moves from input to output without loops
C) Data always loops back
D) Data is never transformed

Answer: B

---

49. Who usually shares responsibility in distributed responsibility frameworks?

A) Only the AI model
B) Only the final user
C) Multiple actors such as developers, firms, and regulators
D) Only the government

Answer: C

---

50. What is one practical purpose of ANN-based finance models?

A) To predict using time-based market patterns
B) To remove all uncertainty from markets
C) To eliminate risk entirely
D) To avoid feature engineering

Answer: A

Online Assessment Questions

1. In AI ethics, what is shared responsibility across actors called?

A) Central accountability
B) Distributed responsibility
C) Legal immunity
D) Individual liability

Answer: B

---

2. What is the use of safeguards to reduce AI harm called?

A) Transfer
B) Defense
C) Acceptance
D) Ignorance

Answer: B

---

3. What is it called when AI risk is shifted to another party?

A) Mitigation
B) Transfer
C) Defense
D) Avoidance

Answer: B

---

4. What does LMS evaluate?

A) Bias only
B) Logic correctness
C) Speed only
D) Cost only

Answer: B

---

5. What does Stereotype Score indicate?

A) Fairness only
B) Bias level
C) Speed only
D) Accuracy only

Answer: B

---

6. What does lack of clear AI accountability create?

A) Efficiency
B) Responsibility gap
C) Transparency
D) Accuracy

Answer: B

---

7. What does mitigation mainly focus on?

A) Eliminating risk fully
B) Reducing impact
C) Transferring risk
D) Ignoring risk

Answer: B

---

8. What does inclusion mainly ensure?

A) Hiring only
B) Participation
C) Salary only
D) Promotion only

Answer: B

---

9. What does diversity mainly focus on?

A) Fairness only
B) Representation
C) Equality only
D) Justice only

Answer: B

---

10. How does systematic disadvantage usually occur?

A) Randomly only
B) Repeatedly across systems
C) Only once
D) Never in practice

Answer: B

Week 12

Concepts

1. Text Mining Basics

Text mining means converting messy text such as reviews, emails, posts, or documents into structured data that a machine can analyze.

Unstructured data is data without a fixed tabular format. Free text is the main example.

The core idea is simple: first structure the text, then analyze it.

2. Lexical and Semantic Approaches

Lexical methods such as Bag-of-Words focus on word presence and frequency but ignore order and deeper meaning.

Semantic analysis tries to understand meaning and context rather than only counting terms.

NLP teaches computers to process human language, while linguistics provides the theory of grammar, meaning, and language structure.

3. POS Tagging and Core Text Terms

POS tagging labels words as noun, verb, adjective, and so on. Nouns often indicate product features or topics, while adjectives often signal sentiment.

A corpus is the full text dataset. A document is one unit inside the corpus, such as one review or one tweet.

A token is a small unit such as a word or symbol. Tokenization is the process of splitting text into tokens.

4. Cleaning and Structuring Text

Stop words such as ?the? and ?is? are often removed because they add little meaning.

Stemming reduces words to a root form, such as turning ?loved? into a stem like ?lov?.

A Term-Document Matrix (TDM) turns text into numbers, where rows represent documents, columns represent terms, and values represent frequency.

n-grams capture word combinations. A bigram has two words, such as ?data mining?.

5. Importance, Similarity, and Sentiment

TF measures term frequency in one document. IDF penalizes terms that appear in many documents. TF-IDF highlights words that are important in one document but not common everywhere.

Cosine similarity compares documents using angle-based similarity. A value closer to 1 means the documents are more similar.

Phi coefficient measures whether words tend to appear together.

Sentiment analysis uses lexicons such as AFINN or NRC to classify language as positive, negative, or emotionally meaningful.

6. R and Tidy Text Concepts

Tidy format means each row is one observation and each column is one variable.

A tibble is an improved data frame in R.

unnest_tokens() performs tokenization. anti_join() helps remove stop words. SnowballC is used for stemming. widyr is used for correlation analysis.

LLMs are advanced language models built on transformers and deep learning. They go beyond simple lexical methods and can model context at scale.

Example: If many movie reviews contain the word ?boring,? TF-IDF may highlight it in a specific negative review, and sentiment analysis may classify the review as negative.

MCQs

1. What is text mining?

A) Data storage only
B) Converting text into structured data for analysis
C) Image processing
D) Networking

Answer: B

---

2. What does unstructured data usually refer to?

A) Tables only
B) Graphs only
C) Free text
D) Numeric vectors only

Answer: C

---

3. What does the Bag-of-Words approach ignore?

A) Words themselves
B) Frequency
C) Context and order
D) Tokens

Answer: C

---

4. What does semantic analysis focus on?

A) Word count only
B) Meaning and context
C) File size only
D) Speed only

Answer: B

---

5. What does NLP stand for?

A) Network Layer Processing
B) Natural Language Processing
C) Neural Logic Programming
D) Numeric Language Parsing

Answer: B

---

6. What does POS tagging identify?

A) Sentences only
B) Grammar roles of words
C) Numbers only
D) Images

Answer: B

---

7. In text mining, nouns often represent what?

A) Actions only
B) Features or topics
C) Errors only
D) Noise only

Answer: B

---

8. In text mining, adjectives often help indicate what?

A) Features only
B) Sentiment
C) Tokens only
D) Document count

Answer: B

---

9. What is a corpus?

A) A single word
B) The full text dataset
C) A token only
D) A line of code

Answer: B

---

10. What is a document in text mining?

A) The whole dataset always
B) One unit of text
C) An algorithm only
D) A matrix only

Answer: B

---

11. What is a token?

A) A paragraph only
B) A chapter only
C) A smallest unit such as a word or symbol
D) A file type only

Answer: C

---

12. What does tokenization mean?

A) Merging text together
B) Splitting text into tokens
C) Deleting text only
D) Sorting files

Answer: B

---

13. What does stemming do?

A) Expands words to full meaning
B) Reduces words to root-like forms
C) Removes all sentiment
D) Detects topics only

Answer: B

---

14. Why are stop words often removed?

A) Because they are the most meaningful words
B) Because they often act as noise
C) Because they are rare
D) Because they are always long

Answer: B

---

15. What does a TDM convert text into?

A) Images
B) Numbers in a matrix form
C) Audio
D) Source code

Answer: B

---

16. In a TDM, what do rows usually represent?

A) Words
B) Documents
C) Tokens only
D) Sentiment scores only

Answer: B

---

17. In a TDM, what do columns usually represent?

A) Documents
B) Words or terms
C) Rows only
D) Sentences only

Answer: B

---

18. What is a bigram?

A) One word
B) A two-word combination
C) A three-word combination
D) A four-word combination

Answer: B

---

19. What does TF mainly measure?

A) Uniqueness only
B) Frequency within a document
C) Document length only
D) Correlation only

Answer: B

---

20. What does IDF mainly measure?

A) Raw term frequency
B) Rarity across documents
C) File size only
D) Token order only

Answer: B

---

21. What does a high TF-IDF value usually mean?

A) The word is common everywhere.
B) The word is important and relatively unique to the document.
C) The word is a stop word.
D) The word is an error only.

Answer: B

---

22. What does cosine similarity equal to 1 suggest?

A) The documents are different.
B) The documents are identical in direction.
C) The documents are random.
D) The documents are empty.

Answer: B

---

23. What does cosine similarity equal to 0 suggest?

A) The documents are identical.
B) The documents are opposite only.
C) The documents are unrelated in direction.
D) The documents are equal in length.

Answer: C

---

24. What does the Phi coefficient measure?

A) File size
B) Correlation or co-occurrence relationship
C) Length only
D) Speed only

Answer: B

---

25. What does sentiment analysis detect?

A) File size
B) Emotional tone or polarity
C) Length only
D) Token count only

Answer: B

---

26. What is AFINN?

A) A clustering algorithm
B) A sentiment lexicon
C) A matrix only
D) A code editor

Answer: B

---

27. What does the tidy approach mean?

A) Messy data layout
B) Structured data layout
C) Random data layout
D) No data layout

Answer: B

---

28. What is a tibble?

A) A graph type
B) An improved data frame in R
C) A token type
D) A word root

Answer: B

---

29. What does unnest_tokens() do?

A) Merges text
B) Splits text into tokens
C) Deletes documents
D) Sorts files

Answer: B

---

30. What does anti_join() often help remove?

A) Tokens entirely
B) Stop words
C) Documents only
D) Rows at random

Answer: B

---

31. What is SnowballC used for?

A) Plotting only
B) Stemming
C) Tokenization only
D) Joining tables only

Answer: B

---

32. What is widyr mainly used for?

A) Plotting only
B) Correlation analysis
C) Storage only
D) Cleaning only

Answer: B

---

33. On what are LLMs mainly based?

A) Sorting only
B) Transformers
C) Tables only
D) Tokens without deep learning

Answer: B

---

34. In which year was the transformer architecture famously introduced?

A) 2000
B) 2010
C) 2017
D) 2022

Answer: C

---

35. What does Word2Vec mainly do?

A) Represents text as numerical vectors
B) Creates images
C) Processes audio only
D) Compiles code

Answer: A

---

36. What does lexical analysis usually ignore?

A) Words themselves
B) Context
C) Tokens
D) Frequency

Answer: B

---

37. What does semantic analysis mainly use?

A) Meaning
B) Count only
C) File size only
D) Token position only

Answer: A

---

38. What can a token be?

A) A word
B) A number
C) A symbol
D) All of these

Answer: D

---

39. Which of the following is an example of a stop word?

A) Data
B) Mining
C) The
D) Model

Answer: C

---

40. What does a corpus contain?

A) Tokens only
B) Documents
C) Words only
D) Numbers only

Answer: B

---

41. The TF part of TF-IDF depends mainly on what?

A) Number of documents only
B) Frequency in the document
C) File size only
D) Time order only

Answer: B

---

42. What does IDF penalize?

A) Rare words
B) Common words across many documents
C) Tokens only
D) Documents only

Answer: B

---

43. What do n-grams help capture?

A) Context between nearby words
B) Noise only
C) File length only
D) Size only

Answer: A

---

44. What does a sentiment lexicon assign?

A) Image labels only
B) Emotional orientation or score
C) File size only
D) Token length only

Answer: B

---

45. What does cosine similarity compare using vectors?

A) Distance by count only
B) Angle between documents
C) Length only
D) Speed only

Answer: B

---

46. What does a Phi coefficient of +1 suggest?

A) No relationship
B) The terms occur together very strongly
C) They never occur together
D) Random behavior only

Answer: B

---

47. What does a Phi coefficient of 0 suggest?

A) No relationship
B) Perfect similarity
C) Exact opposition
D) Equal length only

Answer: A

---

48. At what stage does tokenization occur in the text mining workflow?

A) Data parsing and structuring stage
B) Final visualization only
C) Model deployment only
D) Hardware setup only

Answer: A

---

49. What is the first broad goal in text mining?

A) Plotting results
B) Structuring text for analysis
C) Coding models immediately
D) Deleting noisy data only

Answer: B

---

50. What is the final goal of text mining?

A) Storage only
B) Extracting useful insights
C) Deletion only
D) Formatting only

Answer: B

Online Assessment Questions

1. Why is structuring important in text mining?

A) For storage only
B) For analysis
C) For deletion only
D) For speed only

Answer: B

---

2. Which method mainly ignores context?

A) Semantic analysis
B) Lexical analysis
C) NLP broadly
D) LLM-based reasoning

Answer: B

---

3. What does TF-IDF mainly highlight?

A) Common words across all documents
B) Unique and important words
C) Stop words only
D) Tokens only

Answer: B

---

4. What is a corpus?

A) A single word
B) A text dataset
C) A token only
D) A model only

Answer: B

---

5. What does tokenization split?

A) Numbers only
B) Text into tokens
C) Images only
D) Audio only

Answer: B

---

6. How does stemming help text analysis?

A) It improves normalization of related word forms.
B) It only increases speed.
C) It only increases storage.
D) It only helps plotting.

Answer: A

---

7. What are stop words usually treated as?

A) Highly important terms
B) Noise terms
C) Rare terms only
D) Unique terms only

Answer: B

---

8. What does a TDM convert text into?

A) Tables of numbers
B) Words only
C) Images
D) Sound

Answer: A

---

9. What does cosine similarity compare?

A) Words only
B) Documents or vectors
C) Tokens only
D) File size only

Answer: B

---

10. What does sentiment analysis mainly find?

A) File size
B) Emotion or polarity
C) Length only
D) Token count only

Answer: B